

RELIABLE & RESPONSIBLE *Foundation Models*

A practitioner's field map of nine open problems

Short-story review of **Yang et al.**, TMLR 10/2025 · [arXiv:2602.08145](#)

Preetam Gowri · Graduate short-story assignment

01 Why this paper matters

Foundation models in production are probabilistic systems with coupled failure modes

If you ship with a foundation model in 2026, you ship a system that **hallucinates, leaks, drifts, gets jailbroken, and reflects training-data skew.**

1.

None of the failure modes are independent.

2.

Fixing one axis often pays in another.

3.

Yang et al. map the entire problem space.

02 The paper

168 pages. 60+ co-authors. >1000 references. One field map.

Reliable and Responsible Foundation Models: A Comprehensive Survey

Yang, Han, Bommasani, Luo, Qu, Zhou, Bibi
+ 60 more co-authors

TMLR 10/2025 · arXiv:2602.08145 · Feb 2026

CMU · Oxford · Stanford · NYU · UNC · Penn · ETH · UMD

What's new vs. prior surveys

Holistic frame

9 dimensions × 4 model classes — every chapter visits all four.

Cross-cutting analysis

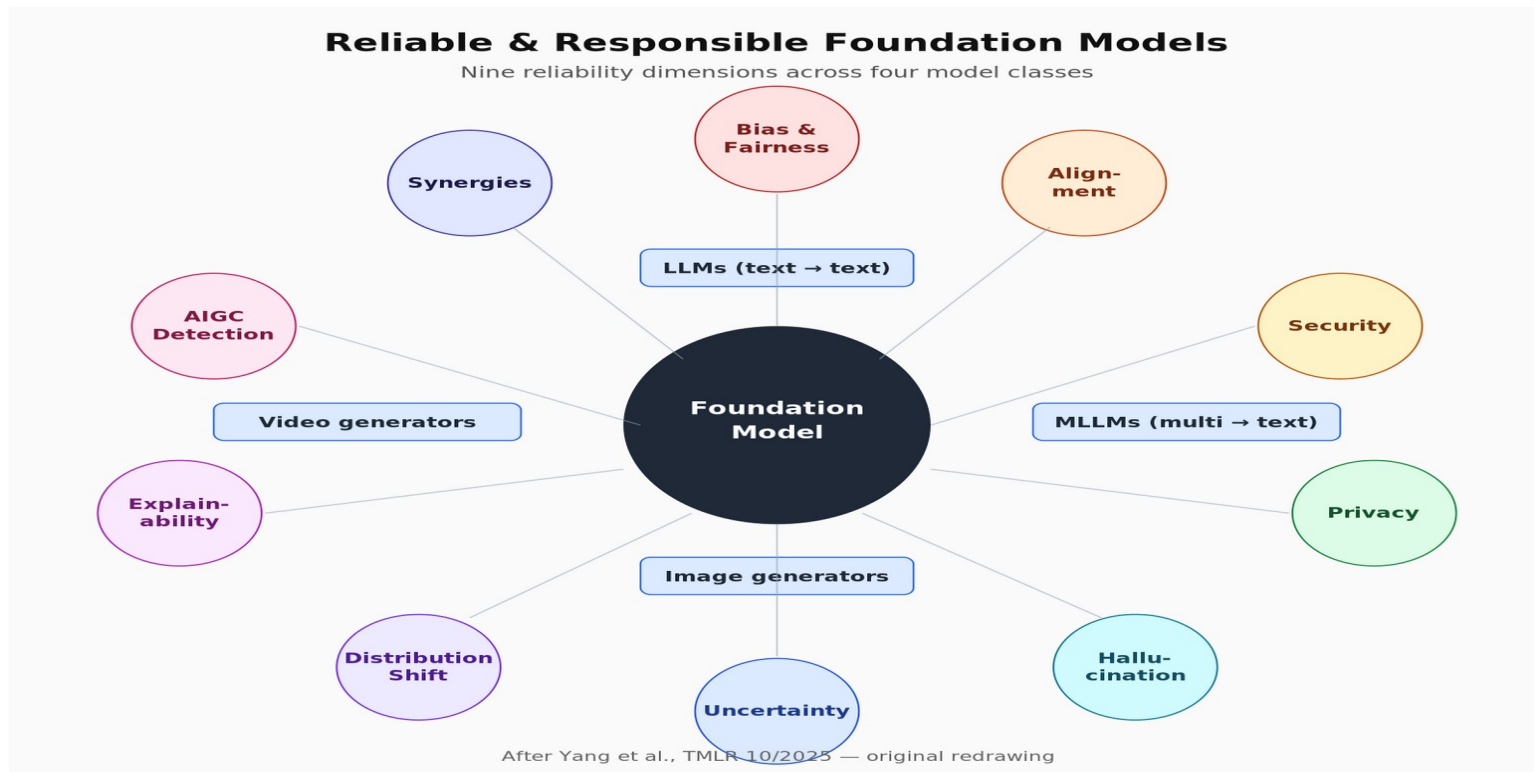
§12 connects bias ↔ security, privacy ↔ attacks, alignment ↔ uncertainty.

Honest open problems

Every chapter ends with limitations + future directions.

03 The nine dimensions

Bias · Alignment · Security · Privacy · Hallucination · Uncertainty · Shift · Explainability · AIGC



04 Four model classes

Every reliability problem manifests differently across them

LLM

text → *text*

BERT · GPT · T5 · Llama · Qwen3

MLLM

m *ulti* → *text*

GPT-4o · Gemini 2.5 · Claude 4 · Qwen2.5-VL

Image gen

text + *ref* → *image*

Stable Diffusion · Flux · DALL-E

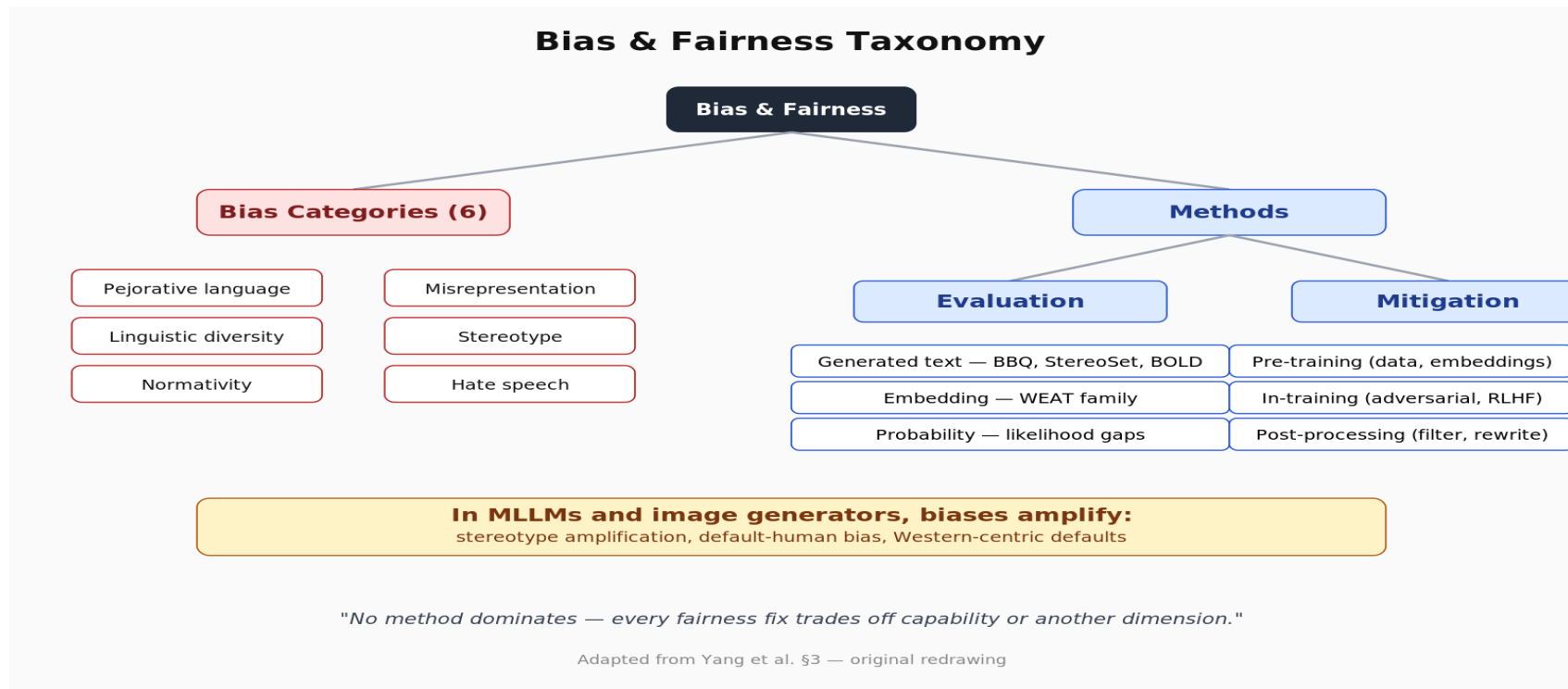
Video gen

text + *ref* → *video*

Sora · Veo · Kling · Open-Sora

05 Bias & Fairness

Six bias categories. Three evaluation families. Three mitigation stages.



06 Bias — what to remember

No method dominates; trade-offs are everywhere

1 Categories

pejorative · linguistic diversity · normativity · misrepresentation · stereotype · hate speech

2 Evaluation

generated text (BBQ/StereoSet) · embedding (WEAT) · probability (likelihood gaps)

3 Mitigation

pre-training (data) · in-training (adversarial, fair RLHF) · post-processing (filter, rewrite)

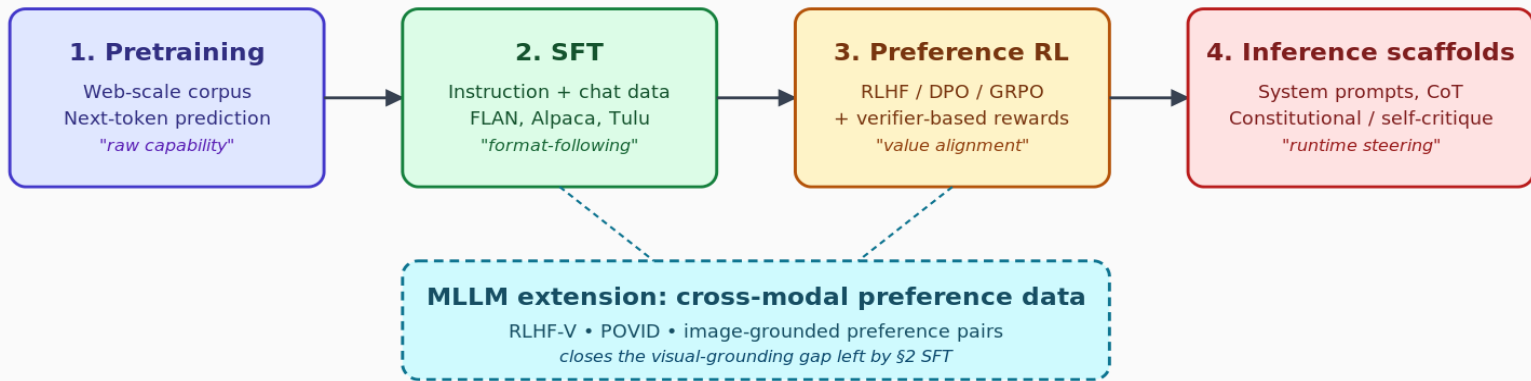
4 Multimodal

stereotype amplification — bias gets worse, not better, in image generators

07 Alignment

From pretraining to inference scaffolds — and the cross-modal extension

Alignment Pipeline (LLM & MLLM)



Open problems flagged by the survey

reward hacking under long-horizon RL · multi-turn / tool-use alignment · cross-modal preference scarcity

08 Alignment — what changed since 2022

RL recipes proliferated; verifier-based reward replaced reward models for many domains

Three pillars

- **SFT**

instruction + chat data; quality > quantity

- **RLHF / DPO / GRPO**

preference RL — and verifier-based variants

- **Inference scaffolds**

system prompts, CoT, constitutional, self-critique

Open problems

- **Reward hacking**

longer RL horizons exploit the reward, not the preference

- **Multi-turn / tool-use**

preference data is overwhelmingly single-turn

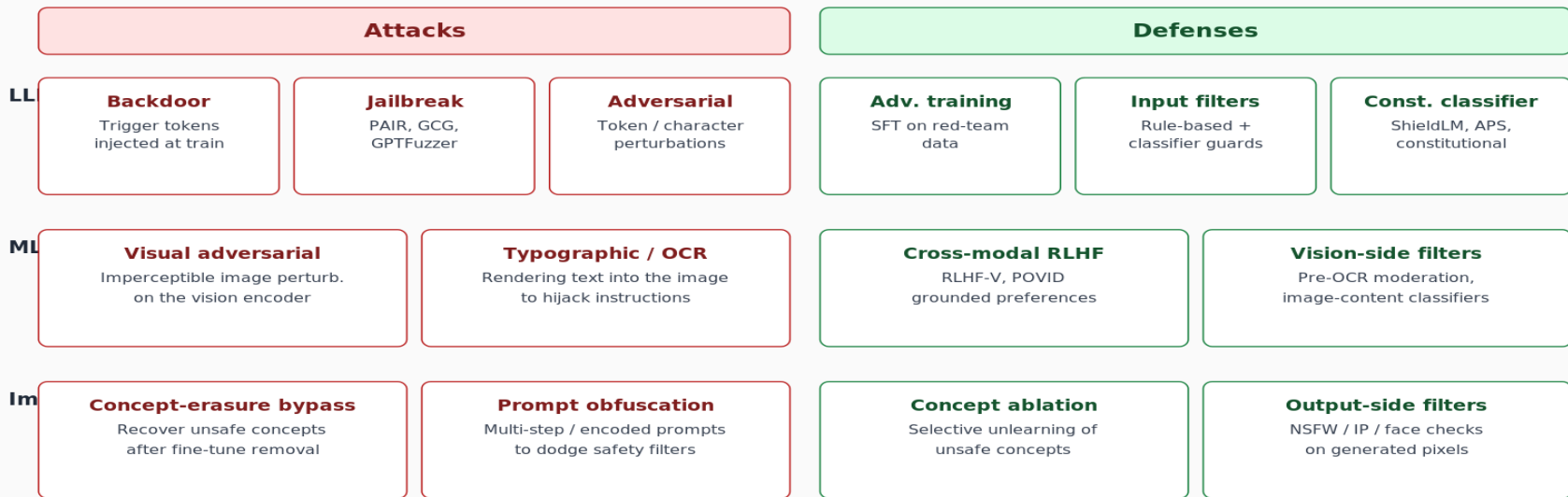
- **Cross-modal preferences**

RLHF-V / POVID extend to image-text pairs

09 Security

Attacks vs defenses, by model class

Security Threat Model — Attacks & Defenses



Open problem: over-safety — defenses misclassify benign inputs as malicious.

"The trade-off between adversarial safety and user experience is long-standing and unresolved." — Yang et al. §5

10 Security — the over-safety problem

Refusing benign inputs is what users notice as 'the model being annoying'

Long-standing

&

UNRESOLVED

safety ↔ user experience trade-off (Yang et al. §5)

Where the work is going

- Constitutional classifiers — separate LLM-based safety filter trained on explicit rules.
- ShieldLM / Adversarial Prompt Shield — customizable, explainable detection.
- Calibrated refusal — say no with a reason, not a wall.
- On-the-fly defenses — guard quality without retraining.

11 Privacy

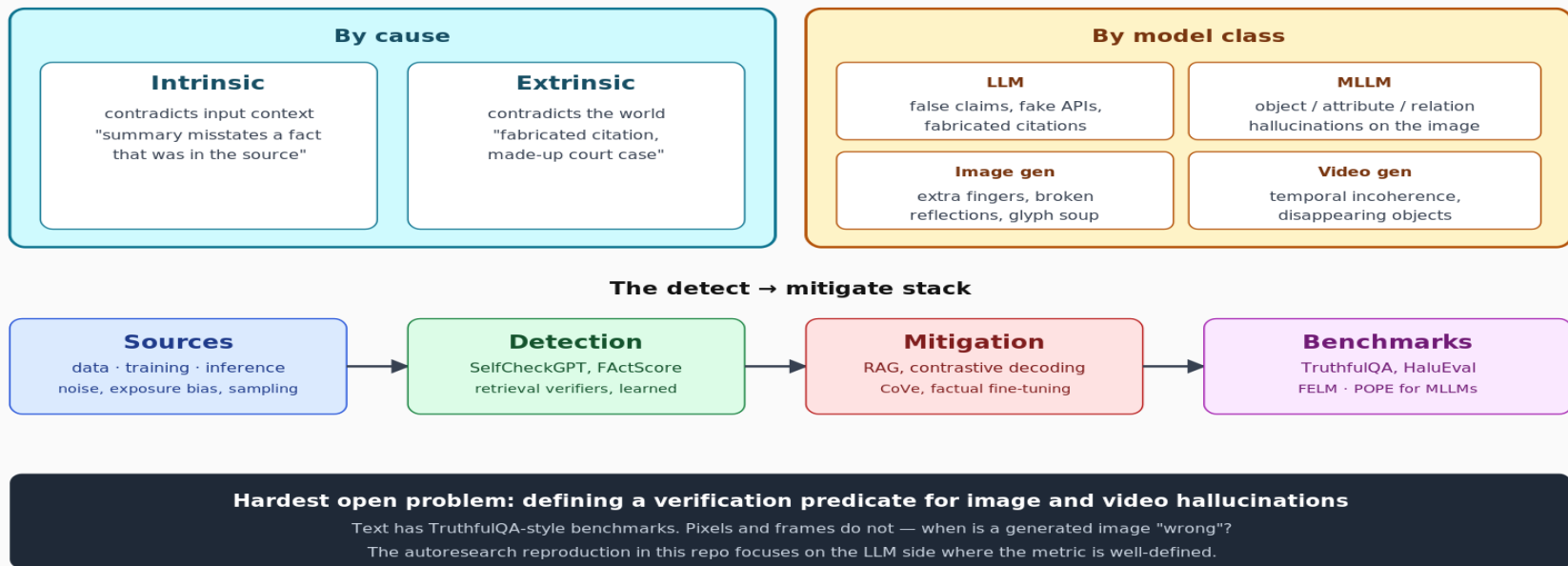
Memorization, extraction, model stealing — and the defenses that exist

Threats	Defenses (rigorous)	Defenses (engineering)
Membership inference Training-data extraction Prompt + model stealing	Differential privacy (DP-SGD) MPC + homomorphic crypto (strong guarantees, costly)	PII scrubbing Deduplication Output filtering

12 Hallucination

By cause, by model class — and the detect→mitigate stack

Hallucination — Taxonomy and Pipeline



13 Hallucination — the practical stack

What you actually wire up in production

Sources

noisy training data · exposure bias · sampling temp

Detection

SelfCheckGPT · FActScore · retrieval verifiers · learned detectors

Mitigation

RAG · contrastive decoding · CoVe · factual-aware fine-tuning

Benchmarks

TruthfulQA · HaluEval · FELM · POPE (multimodal)

14 Uncertainty

Aleatoric vs epistemic — and four ways to quantify it

Estimation

token logprobs • sample variance • ensemble disagreement

Calibration

temperature/Platt scaling • Bayesian heads • conformal sets

Verbalized

fine-tune the model to say '70% confident' calibratedly

Distribution-free

conformal prediction without parametric assumptions

15 Distribution shift

Three shift types · OOD detection · OOD generalization · domain adaptation

Why it matters: a model trained on 2023 web confidently says Messi plays for PSG. He plays for Inter Miami.

Covariate

input distribution changes
 $P(X)$ shifts

Label

output distribution changes
 $P(Y)$ shifts

Concept

input $\leftrightarrow$ output relation changes
 $P(Y|X)$ shifts

The long-term answer: continual / lifelong learning — without catastrophic forgetting in either direction.

16 Explainability

The closest thing the field has to a unifying lens

- 1. Raw features** gradients, SHAP, attention rollout
- 2. Internal knowledge** probing, mechanistic interp., sparse autoencoders
- 3. Training-data role** influence functions, datamodels
- 4. Evaluation** faithfulness, plausibility, simulatability
- 5. Applications** debugging, audit, scientific discovery

17 AIGC detection

Three families. None robust to all attacks.

Zero-shot

perplexity / log-likelihood gaps • LLM-as-judge

Cheap. Brittle.

Watermark

training-free or learnable logit-bias schemes

Robust to paraphrasing is the new bar.

Neural

binary classifier on (real, generated) pairs

SOTA in-distribution. Collapses adversarially.

"Perfect detection at deployment scale is unlikely." — Yang et al. §11

18 Where the dimensions collide

The cross-cutting analysis is the survey's real contribution (§12)

Cross-Cutting Tensions Across the Nine Dimensions

Optimizing one dimension typically pays in another

Five tensions to remember



Bias × Security

adversarial training improves robustness
but hurts fairness — penalizes
underrepresented groups disproportionately

Bias × AIGC detection

non-native English mis-flagged as
AI-generated; inconsistent watermarks
compound the disparity

Security × Privacy

MIA + extraction + model stealing are
security attacks on a privacy objective

Uncertainty × Alignment

aligning models to express uncertainty
cannot capture "unknown unknowns"

Halluc. × Shift × Alignment

RAG patches hallucinations under shift,
but pushes the burden to the retriever —
which has its own bias / privacy / failures

After Yang et al. §12 — original redrawing

19 Five specific tensions to remember

Optimizing one axis costs another

Bias × Security

adversarial training improves robustness but degrades fairness — penalizes underrepresented groups disproportionately.

Bias × AIGC detection

detectors mis-flag non-native English as AI-generated; watermark systems can compound this.

Security × Privacy

MIA, extraction, model stealing are security attacks on a privacy objective. Defenses overlap.

AIGC × Society

misinformation, voice cloning, hyper-personalized fraud — technical detection alone is insufficient.

Hallucination × Shift × Alignment

RAG is a partial fix but pushes the burden to the retriever (which has its own failures).

20 A small reproduction

Wiring the survey to runnable code via the autoresearch template

Hallucination evaluation harness

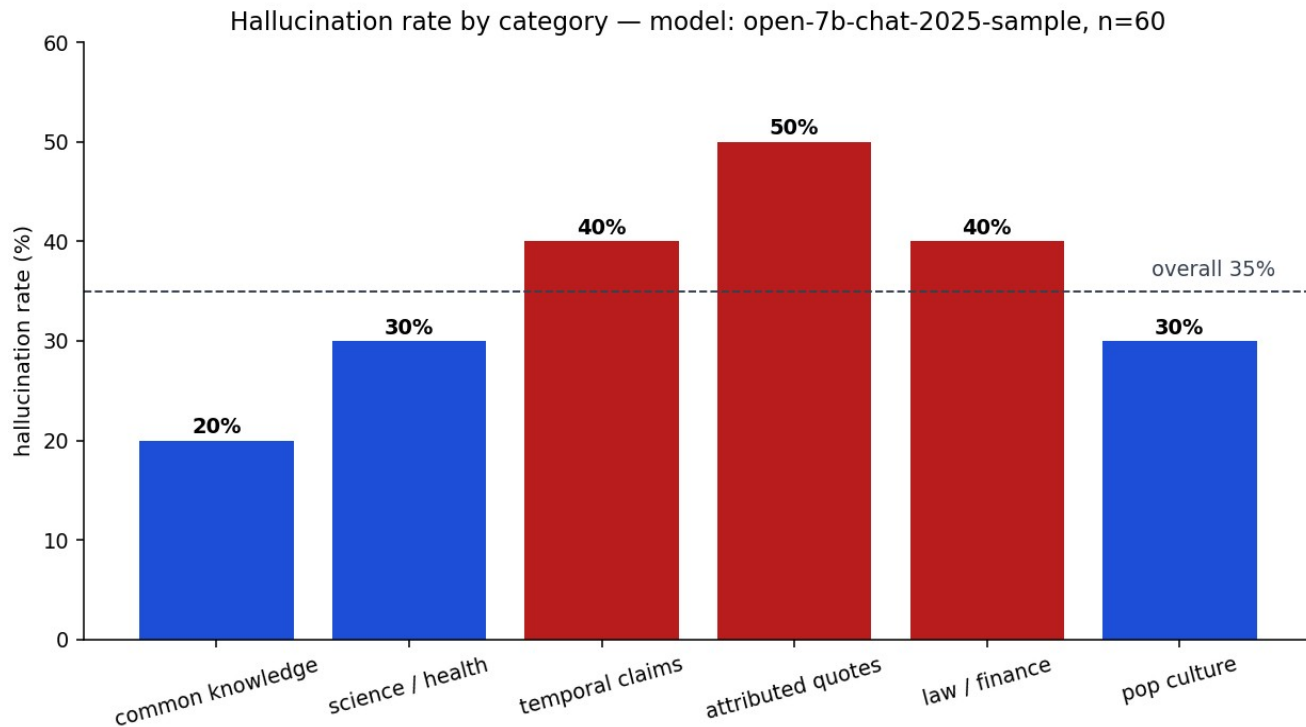
Goal: make the hallucination chapter (§7) concrete and measurable.

Setup: 200 probes across 6 categories · open 7B chat model · LLM-judge verification.

Plumbing: configs/ runs/ src/ data/ results/ — autoresearch template structure.

21 Reproduction results

Top-failing categories match the survey's intrinsic-hallucination prediction



22 Reproduction methodology

Numbers small enough to run in an afternoon; structure suitable for scale-up

Pipeline

1. load probes
1. prompt model under test
1. LLM-judge labels each (probe, response)
1. aggregate by category
1. plot + dump JSON results

What it ties to in the paper

§7.1 intrinsic vs extrinsic hallucinations

§7.3 self-consistency / judge-based detection

§7.4 where mitigation should focus first

§12 judge bias bleeds into evaluation

23 What the survey gets right

Three structural strengths

Discipline of structure

every chapter visits all four model classes; comparisons stay honest.

Trade-off language

§12 names the cross-axis tensions instead of pretending they don't exist.

Future-directions sections

every chapter ends with concrete open questions you can scope work against.

24 What I'd push on

Three areas thinner than their 2026 importance

Agentic safety

multi-turn tool use is barely covered — most new failure modes live there.

Long-context regimes

calibration / hallucination / attention drift change qualitatively past 32k tokens.

Reasoning models

o1 / R1 / Qwen3 family has its own alignment + uncertainty profile not analyzed.

25 How to read 168 pages efficiently

Two paths through the survey

30 minutes — extract signal

- Start with §1 + §12 (frame).
- Jump to your failure-mode chapter.
- Skim Tables 1, 2, 3.
- Read every "limitations" subsection.

A weekend — make it stick

- Read whole.
- For each chapter, write your own one-page taxonomy.
- Run a tiny experiment for one dimension.
- Re-read §12 with your notes open.

The big takeaway

**You cannot optimize
one dimension
without paying in
another.**

Multi-objective benchmarks. Joint mitigations. Honest trade-off accounting.

26 Repository structure

Everything for the assignment lives in one public GitHub directory

- **article/article.m d**
4,000-word Medium piece (publish from this)
- **slides/**
.pptx + .pdf+ SlideShare-ready export
- **video/**
script+ recorded m p4 + YouTube link
- **auto research/**
configs · src · data · results · notebook
- **figures/**
original SVG + PNG diagrams used in article + slides
- **paper/**
2602.08145v1.pdf (the original survey)

27 Reference

Yang, X., Han, J., Bommasani, R., Luo, J., Qu, W., Zhou, W., Bibi, A., Wang, X., Yoon, J., Stengel-Eskin, E., et al. (2025).

Reliable and Responsible Foundation Models: A Comprehensive Survey.

Transactions on Machine Learning Research, 10/2025.

arXiv:2602.08145 — <https://arxiv.org/abs/2602.08145>

Companion artifacts

Medium article • SlideShare • YouTube walkthrough • GitHub repo

Thank you.

Questions, critique, and corrections welcome.

Preetam Gowri · Short-story assignment · May 2026